

Client Identity Matching

Frontend change report - unified client profiles from duplicate rows

1. Overview

Multiple dashboard rows can represent the same real-world client when core identity fields align. This feature treats those rows as one client: clicking any matching row opens the same profile, and the profile aggregates record-specific data (years, rep, calculations, calls, added dates) without merge buttons or new screens. Implementation is frontend-only with minimal, localized changes.

2. Feature Location

- Client Dashboard -> All Leads & Clients table (Status column / row click)
- Client Profile page (/clients/\$id)

3. Matching Rules (ALL required)

- Client name: fullName must match exactly (strict equality).
- Entity: at least one entity name must match exactly (from entityNames or data.entities).
- Contact: normalized phone matches OR normalized email matches (non-empty).

Clusters use transitive matching: if record A matches B and B matches C, all three belong to the same client group even when A and C do not match directly.

4. Canonical Profile ID

Within a cluster, the canonical lead id is the record with the earliest addedAt date; ties break on lowest numeric id. Pipeline navigation always routes to this id. Profile mutations (edit, notes, tax history) target the canonical record.

5. Files Changed

- frontend/src/utils/clientIdentity.ts - NEW - matching, clustering, merge utilities
- frontend/src/utils/clientIdentity.test.ts - NEW - 6 unit tests
- frontend/src/pages/PipelinePage.tsx - Navigate to canonical profile id
- frontend/src/pages/ProfilePage.tsx - Cluster load, aggregation, unified display

Note: frontend/package-lock.json may show unrelated local drift and is not part of this feature.

6. New Utility: clientIdentity.ts

Exported functions

```
entityNamesFor(lead) - collect entity names from list + data
leadsMatch(a, b) - pairwise match on name + entity + contact
getClientCluster(lead, allLeads) - BFS transitive cluster
canonicalLeadId(lead, allLeads) - stable profile id for navigation
mergeLeadsForDisplay(cluster, canonicalId) - merged read model
clientRecordSnapshots(cluster) - per-record field snapshots
```

Contact normalization

```
Phone: strip non-digits, compare digit strings
Email: trim + lowercase, compare (must be non-empty)
```

frontend/src/pages/PipelinePage.tsx

Client Dashboard table row click handler.

- Import canonicalLeadId from @/utils/clientIdentity.
- goProfile() resolves canonicalLeadId(l, leads) before navigate.
- All matching rows open /clients/{canonicalId}.
- goCalc() unchanged - still uses the clicked row's lead for calculator context.

frontend/src/pages/ProfilePage.tsx

Client Profile - load, display, and mutate unified client.

- routeLead: lead loaded by URL id; cluster via getClientCluster.
- canonicalId: stable id for writes and sub-navigation.
- lead: mergeLeadsForDisplay when cluster size > 1.
- primaryLead: canonical record used for EditClientDialog.
- Fetch all cluster members (fetchLead, fetchCalls, fetchIntakeNotes).
- Merge calls and intake notes from all cluster lead ids.
- updateLead / intake notes use canonicalId.
- Call toggle/auto-complete resolves owning lead id per call.
- Feasibility Call and ScheduleCallDialog use canonicalId.

7. Profile Aggregation (existing UI only)

Merged automatically on display lead

- taxYears - union of all cluster records, sorted
- entityNames / entitiesCount - combined
- engagements - concatenated from all records
- latestCalculation - most recent date across records
- calls - all follow-up calls from cluster members
- intakeNotes - all notes, newest first

General Information (when cluster has 2+ records)

- Engagement Years (all records)
- Representatives (all records)
- Latest Calculations (all records)
- Next Calls (all records)
- Added Dates (all records)

Assigned Sales Representative card

- Shows comma-separated list when multiple reps exist in the cluster.

8. Expected Behavior

- Duplicate rows with matching name, entity, and phone/email open the same profile.
- No merge button, merge workflow, or new screens added.
- Record-specific data remains visible in aggregated fields.
- Single-record clients behave exactly as before.
- Backend APIs unchanged; no schema or server logic modified.

9. Unit Tests (clientIdentity.test.ts)

- Matches when name, entity, and email align.
- Rejects match when entity differs.
- Rejects match when contact differs.
- Clusters transitively through a middle record.
- Canonical id prefers earliest addedAt.
- mergeLeadsForDisplay combines tax years and latest calculation.

10. Validation Checklist

- Find duplicate rows: same fullName, shared entity, shared phone or email.
- Click Status/name from different matching rows -> same profile URL.

- Verify aggregated years, reps, calculations, calls, added dates on profile.
- Confirm filters, search, and single-client workflows still work.

11. Restrictions Followed

- No push to GitHub.
- No commits created as part of this report.
- Frontend-only; minimal non-invasive changes.
- No new UI components, merge buttons, or screens.

12. Git Status (at report time)

Branch: Year-Range-Fix-2022-2026. Client matching changes are local and unstaged: 2 new files, 2 modified source files. Regenerate this PDF after further edits with: `python scripts/generate_client_identity_changes_pdf.py`